# Project Design Phase-II
# Data Flow Diagram & User Stories

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|---|---|
| Date | 31 January 2026 |
| Team ID | [Enter Team ID] |
| Project Name | Global Fuel Prices and Taxation Analysis (2020 to 2026) |
| Maximum Marks | 4 Marks |

**Data Flow Diagrams:**

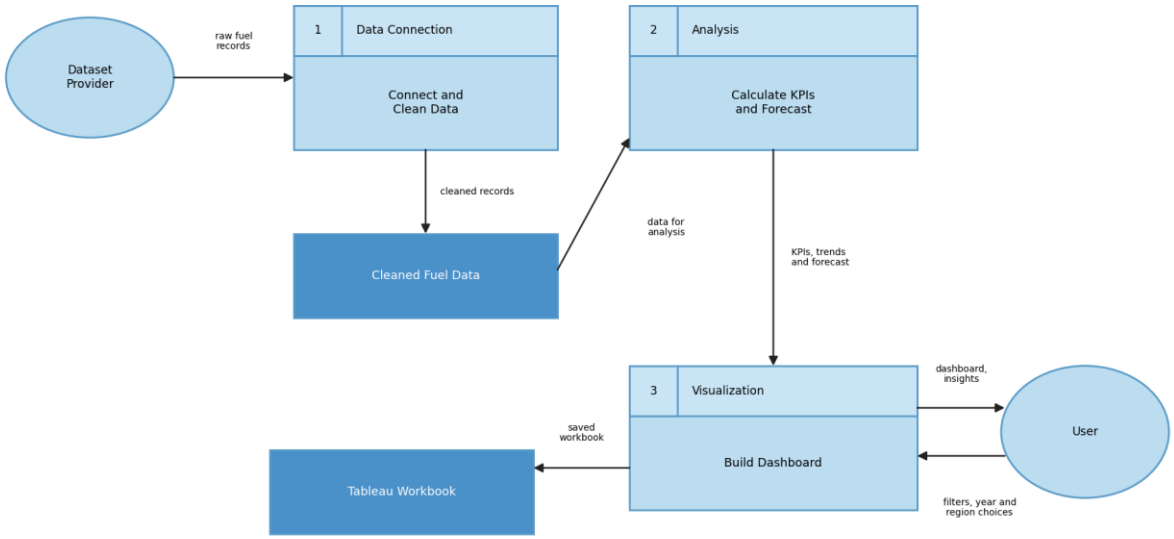
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

**DFD Symbol Legend**

| Symbol | Name | Description |
|---|---|---|
| **Oval / Rounded shape** | **External Entity** | A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway). |
| **Rectangle with numbered header** | **Process** | An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order). |
| **Rectangle (solid fill, no number)** | **Data Store** | A place where data is held for later use (e.g., Customer Database, Inventory). |

| | | |
|---|---|---|
| **Labeled arrow** | **Data Flow** | The movement of data between entities, processes, and data stores, labeled with what data is being passed. |

DFD Level 0: Global Fuel Prices and Taxation



## User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Policy Consultant (Dr. Mitchell) | Regional Comparison | USN-1 | As a policy consultant, I can view a map of average fuel tax by country so that I can see which countries tax fuel the most. | Map colors every country by average tax %; tooltip shows region and value | High | Sprint-1 |
| | Tax and Subsidy | USN-2 | As a policy consultant, I can compare average tax by region and subsidy level so that I can spot policy gaps. | Chart shows tax % for each region, split by subsidy level | High | Sprint-1 |
| | Subsidy Impact | USN-3 | As a policy consultant, I can compare fuel price across subsidy levels so that I can see how subsidies affect price. | Chart shows average price for each subsidy level and region | High | Sprint-1 |
| Supply Chain Manager (Mr. Thompson) | Price Forecast | USN-4 | As a supply chain manager, I can see diesel price trends with a forecast so that I can plan fuel purchases. | Line chart shows monthly diesel price with a forecast period | High | Sprint-1 |
| | Market Ranking | USN-5 | As a supply chain manager, I can see the 10 most and 10 least expensive countries so that I can choose favorable markets. | Two ranked bar charts show top and bottom 10 countries by average price | High | Sprint-1 |
| | Price Risk | USN-6 | As a supply chain manager, I can see price volatility by region and year so that I can find unstable markets. | Heatmap shows volatility by region and year; darker means less stable | Medium | Sprint-2 |
| Researcher (Prof. Rodriguez) | Income Analysis: Tax | USN-7 | As a researcher, I can compare average tax across income levels so that I can test whether richer countries tax fuel more. | Bar chart shows average tax % for each income level | High | Sprint-2 |
| | Income Analysis: Subsidy | USN-8 | As a researcher, I can see the share of each subsidy level within each income group so that I can judge affordability support. | Chart shows subsidy level share for each income level | High | Sprint-2 |

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|---|---|---|---|---|---|---|
| | Income Analysis: Price | USN-9 | As a researcher, I can compare petrol, diesel and LPG prices by income level so that I can study affordability. | View shows average price of each fuel type for each income level | Medium | Sprint-2 |
| All Users | Filtering | USN-10 | As a user, I can filter all charts by year so that I can look at one period. | Choosing a year updates all nine charts | Medium | Sprint-3 |
| | Click-to-Filter | USN-11 | As a user, I can click a country or region on the map so that the other charts focus on it. | Selecting a mark filters the other linked sheets | Medium | Sprint-3 |
| Administrator | Reporting | USN-12 | As an administrator, I can refresh the data and export the dashboard as an image or PDF so that I can share results. | Refresh updates all sheets; exported file opens correctly | Low | Sprint-3 |